

CIS129 Project 1

The purpose of this project is to test some of the scripting knowledge that you have learned or will learn before the Midterm Exam. Your task is to find the solution (Unix command(s)) to each of the items below, then put your solution into a file, execute a shell to run your commands and capture the output into another file.

You will: (6 Points)

1. Put all your commands for this project in one text file: **project1.sh**
2. Find solutions to the items posed below.
3. Put a comment on the line above each of the item sections, which explains the command(s) you are using.

For example, if the item is to find the number of users on the system:

```
# Number of users on the system using the who command.
echo "The number of users currently logged in is: "
who -q
echo
echo
```

OR

```
# Number of users on the system using the who and wc commands.
echo "The number of users logged in is: `who | wc -l`"
echo
echo
```

Both above examples would be a valid answer for that question.

IMPORTANT! Notice how the examples above label the output so that the user can tell what they are looking at and reading! **EACH** item below **MUST** have labels!

IMPORTANT! Make sure that you put blank lines in the script for easier reading and debugging as well as in the output!

A couple of script requirements: (6 Points)

1. You should use the bash shell to complete this project.
2. **You must use a function to create the student header information** that uses "echo" or printf statements to complete the heading; project number, your name, date/time. Notice that the date will show the current date/time the script was executed.

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Example heading: **MAKE SURE YOU PUT IT IN A FUNCTION!**

```
echo "*****"
echo
echo "CIS 129 Project 1"
echo "Your Name"
echo `date`
echo
echo "*****"
```

Find solutions to the following items: (4 Points Each)

Remember to refer to the **IMPORTANT** items above for each of the following!

- a) Show information about your UID, GID, and any groups you belong to.
- b) Show your UNIX login name.
- c) Show the number of users currently logged in.
- d) Create an alias called `n` that runs the command in item c above. Then use the new alias `n` to show the number of users currently logged in.

NOTE: You may need to use the `shopt` command for the alias to work. After you create the alias, you will need to use the `shopt` command to export the alias into your current session **BEFORE** you try to use the alias.

```
shopt -s expand_aliases
```

- e) Show the directory listing of your home directory in reverse alphabetical order.
- f) Show the absolute path to, and the name of, your home directory.
- g) Show the last time the cislinux server was booted.
- h) Show the monthly calendar for the month/year you were born.
- i) Show the monthly calendar for the month/year you were born BUT translate every 1 to an a, every 2 to a b and every 3 to a c.
- j) Using the paste command, display the 2 calendars from items h and i above next to each other.
- k) Using the date command and the cut command and piping, display just the current 3 character month.
- l) Show the first 3 lines of the `/etc/passwd` file.
- m) Show the first 3 lines of the `/etc/passwd` file with each line proceeded by:
 `"#: ** > "` where the # is the line number.
- n) Find the number of characters in the password file, on the server. `/etc/passwd`
- o) Show the first 5 lines of the `/etc/group` file in Hexadecimal.
- p) Show the last 15 users that logged into the cislinux server.

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- q) Show the last 15 users that logged into the cislinux server sorted in REVERSE alphabetical order.
- r) Using Command Substitution, Environment Variables, and an echo command, show your username and default shell as follows:

EXAMPLE: My username is: brent. My shell is: /bin/bash

- s) Create a directory called MICHIGAN.
- t) Copy all the script files that end in .sh from **your home directory** into the MICHIGAN directory.

IMPORTANT: Do **NOT** use the -r switch on the cp command when copying the files. It is **NOT** required and will cause other directory structures to be copied and then they cannot be removed!

- u) Show the directory listing for the MICHIGAN directory in reverse alphabetical order.
- v) Remove the MICHIGAN directory and all files in that directory **AND** exit the script normally.

Deliverables

- a) Run your script file and correct any errors that are displayed.
RUN: `bash project1.sh`
- b) Once you have corrected the errors, redirect the output to a file.
RUN: `bash project1.sh > project1.out`
- c) Upload **both**: **project1.sh AND project1.out** files into Moodle